

Safe Alone, Risky Together

A Visual Tool for Understanding Over-the-Counter Drug Risks

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Abstract

Over-the-counter (OTC) drugs are fundamental to self-care in the United States, utilized by almost 90% of persons and frequently selected without medical advice. Despite their commonplace nature, these goods harbor concealed risks—especially when various medications contain identical active components, resulting in possible overdose or harmful interactions. Despite the availability of safety information on standardized “Drug Facts” labels, research indicates that numerous consumers either neglect or find it challenging to comprehend these labels due to inadequate health literacy, diminutive font sizes, and the complex terminology of pharmaceutical language. This design thesis tackles the gap by presenting a user-centered, interactive visualization tool that assists consumers in identifying overlapping chemicals and comprehending acceptable dosage thresholds for common OTC drugs.

The tool, constructed with Vue.js and D3.js, comprises six successive sections—Problem, Ingredients, Categories, Try, Dosage Guide, and Label Check—each aimed at facilitating users' progression from basic awareness to more informed and contemplative decision-making. By redirecting attention from brand familiarity to ingredient comprehension, the interface prompts users to reflect and inquire: what substances are actually present in the medication I am consuming? The application converts intricate pharmaceutical data into clear, actionable insights using visual components like pill icons, arc diagrams, and color-coded compounds. Users can simulate combinations of up to three OTC medications and receive immediate alerts on overlapping ingredients or hazardous dosage levels, so aiding in the prevention of inadvertent usage.

This project reconceptualized pharmaceutical safety as a design challenge rather than only a medical one, intersecting with public health, digital literacy, and human behavior. It illustrates

how visual storytelling and engagement may connect access and comprehension, particularly for individuals with constrained health literacy. As healthcare systems progressively impose decision-making responsibilities on individuals, this technology provides a scalable, visual-centric framework for conveying essential medication safety information at the point of decision. The research culminates with prospective enhancements, encompassing prescription medicine integration, machine learning-based picture analysis, and mobile-compatible, multilingual accessibility.

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1. Introduction

1.1 Background and Motivation

Over-the-counter (OTC) medicine, also known as nonprescription medicine, refers to a broad category of drugs that can be purchased without a prescription. These include a wide range of products used to treat everyday symptoms such as headaches, colds, allergies, and indigestion. For most Americans, they are not only familiar but essential: 9 out of 10 people use OTC medications regularly, and 8 out of 10 do so without consulting a healthcare professional【19】. With nearly 2.9 billion OTC purchases each year in the U.S., these products have become a routine part of everyday health maintenance. Their widespread availability—in pharmacies, convenience stores, vending machines, hotel pantries, and online platforms—further reinforces the public’s perception that OTC drugs are safe, simple, and low-risk【19】.

1.2 Problem Statement

But that perception is incomplete. In designing this project, I was struck by how little most people (myself included) know about what’s actually inside the medications we casually rely on. I began noticing how often I—and people around me—combine products without checking if they contain the same active ingredients. Many of these medications overlap in function and composition, and when taken together, they can easily lead to unintentional overdoses or adverse effects like liver damage【5】【18】. According to the World Health Organization, more than 50% of all medications globally are used incorrectly—suggesting a widespread gap between access and understanding【18】.

This issue is especially pressing in the U.S., where high healthcare costs and long wait times have made self-medication the norm. 81% of American adults turn to OTC medications as

their first response to minor symptoms【5】. Yet studies show that only half of them read the Drug Facts label, and just 77.3% know how to interpret the safety information it provides【18】. On average, people visit a store 26 times a year to buy OTC medications—but see a doctor only three times【4】. The disconnect between how often these drugs are used and how little people know about them is what motivated this work.

1.3 Research Questions and Design Direction

I started asking myself—how can something so common be so misunderstood? The answer, in part, lies in how the information is presented. Labels are technical, dense, and hard to read. Warnings are often printed in small fonts, buried in packaging. Even when people want to make safe decisions, the format doesn't help them. That realization made this not just a public health concern, but a design problem.

As someone working at the intersection of design and data, I saw an opportunity to use visualization not just to explain information, but to reframe it. Instead of abstract tables and blocks of text, what if people could see what ingredients show up across products? What if they could simulate combining cold medicine and a painkiller, and immediately get a warning if those drugs share the same chemical? What if dosage information wasn't hidden in milligrams, but shown visually, as a series of pills with safe limits clearly marked?

This project emerges from that intention. It aims to make the hidden risks of OTC medications more visible—especially the overlap of active ingredients and the potential consequences of misusing them. I've designed an interactive tool that invites users to explore ingredient data visually and intuitively. The goal is not just to deliver information, but to

nudge behavior—to help people pause before they reach for a second bottle, and ask: what’s really inside this one?

1.4 Guiding Questions

This project is driven by questions that are both design-based and personal:

- How can we help users recognize when different medications contain the same active ingredients?
- Can a visual format communicate risks and dosage more effectively than traditional labeling?
- What kind of design encourages people to pause, reflect, and choose more consciously—not just based on brand, but based on what’s inside?

To fully understand why so many Americans rely on OTC medications in the first place—and why those design questions even matter—we need to consider the broader healthcare environment in which these decisions take place.

2. Related Work

2.1 OTC Usage and Self-Care Culture

While this project focuses on the micro-level decisions consumers make at the pharmacy shelf, those choices are inextricably linked to the macro-level conditions of the American healthcare system—where structural barriers, high costs, and fragmented access have made self-care not just a preference, but in many cases, a necessity. The United States is one of the highest-spending countries in the world when it comes to healthcare, with total expenditures reaching approximately \$4.9 trillion in 2023—translating to around \$14,570 per person and

accounting for nearly 17% of the nation's GDP [3]. Despite this massive financial outlay, the American healthcare system is marked by limited public coverage and a fragmented insurance landscape, leaving many individuals either uninsured or underinsured. Most health insurance is tied to employment or must be purchased independently, while public options like Medicare and Medicaid are reserved for specific populations such as the elderly and low-income individuals[6]. As a result, around 9% of the population remains uninsured, and even those with coverage often face prohibitively high out-of-pocket costs[3].

This structural and financial reality has significantly contributed to the normalization of self-medication in the U.S. Roughly one in four American adults report having delayed or foregone needed medical care due to cost, and among the uninsured, that figure rises to over 60%[6]. Access barriers don't stop at cost—wait times are another major issue. A 2023 national survey conducted by the American Association of Nurse Practitioners (AANP) found that more than 40% of respondents described wait times as “significantly long,” and 26% had to wait more than two months to see a provider. Many ended up canceling their appointments altogether, including those seeking essential mental health services[2]. Within this landscape, where affordability and timely access are far from guaranteed, it's no surprise that many Americans turn to self-care for minor symptoms rather than navigating a slow and costly healthcare system.

Against this backdrop, over-the-counter (OTC) medications have become a practical, accessible alternative to formal medical care. For a wide range of everyday health concerns—such as colds, headaches, allergies, or indigestion—93% of American adults report choosing OTC products over visiting a doctor or clinic[5]. This reflects not only trust in the efficacy of OTC drugs, but also a broader desire to avoid the time, cost, and complexity of professional medical intervention. In an overburdened healthcare system, where wait times

can be long and appointment access uneven, OTC medications offer an immediate, self-directed solution.

The benefits extend far beyond the individual. For every dollar consumers spend on OTC medicine, the U.S. healthcare system is estimated to save six to seven dollars in avoided clinical visits, prescription costs, and diagnostic procedures. These savings add up: they ease the load on healthcare providers and reduce insurance spending, while also enabling patients to manage symptoms without disrupting their daily routines. An estimated 61 million American workers are able to treat minor ailments without missing work or scheduling medical appointments, contributing directly to workplace productivity and economic continuity[5].

This constellation of economic and logistical advantages has helped cement OTC drugs not merely as consumer goods, but as essential tools in America's self-care ecosystem. Their continued growth in usage reflects more than convenience—it reveals a distinct cultural orientation toward health itself.

Underlying this trend is a deeper cultural framework rooted in American ideals of autonomy and self-determination. In this context, health is not always approached as a public good or collective right, but often as a matter of personal responsibility and private decision-making. The dominant neoliberal logic reframes healthcare as a consumer marketplace, where individuals are expected to shop for services, weigh risks, and make cost-effective choices. Medical decisions thus become acts of independence, and patients are recast as consumers managing their own wellness portfolios.

Within this framework, OTC medications emerge as empowering tools—products that allow individuals to bypass institutional gatekeepers and take charge of their health on their own terms. Their easy availability in non-clinical spaces like grocery stores and gas stations blurs

the boundary between medical care and everyday consumption. This normalization reinforces a worldview in which medical needs are expected to be met quickly, affordably, and without outside interference. As a result, OTC medications play an increasingly central role in shaping not only how Americans treat illness, but how they define and perform health itself. But as these products take on a larger role in everyday health management, the responsibility of understanding what's actually in them has shifted almost entirely onto the consumer. The effectiveness of this self-directed care model depends heavily on the clarity and usability of the information provided—and that's where many existing systems begin to show their limitations.

2.2 Drug Facts Label Limitations

Since 1999, the U.S. Food and Drug Administration (FDA) has required all over-the-counter (OTC) medications to display a standardized “Drug Facts” label to help consumers choose and use these products more safely. This label includes information such as:

- **Active ingredients and purposes** (the main components of the drug and what they are used for),
- **Warnings** (potential side effects, allergic reactions, use during pregnancy, etc.),
- **Usage instructions** (dosage, frequency, and duration), and
- **Other details** (storage instructions, inactive ingredients, etc.)【14】.

However, the regulation sets a minimum font size of just 6 points, and since a significant amount of content must be included in a limited package space, the resulting layout tends to be densely packed and difficult to read. Research indicates that adults over the age of 60 typically require at least a 6.7-point font for legibility【20】. This suggests that older adults or

those with visual impairments may struggle to access the information on these labels effectively.

In addition, many OTC drug labels are written at a 7th-grade reading level or higher, with some requiring comprehension at a 9th-grade level or above【16】. For users with low health literacy, this can pose a major barrier to understanding, especially when faced with unfamiliar terminology or complex ingredient names. Moreover, studies have shown that among 100 analyzed OTC drug labels, approximately 90 to 96 lacked sufficient or complete side effect information, pointing to the need for clearer and more standardized guidance【4】.

2.3 Online Drug Information Platform Limitations

WebMD and Drugs.com are among the most widely used online health information platforms in the United States, providing consumers with tools to search for symptoms, check drug interactions, and access detailed medication information. WebMD offers a comprehensive health portal that includes condition overviews and interactive tools, while Drugs.com focuses more specifically on both prescription and OTC drugs, offering detailed explanations, side effect listings, and user reviews.

Despite their popularity, these platforms often present information in ways that can be difficult for general users to navigate. Most content is organized around clinical terminology and long-form text, which can be overwhelming—especially for users unfamiliar with drug ingredient names. Even those who frequently use OTC products may recognize the package design more readily than the active ingredients, and needing to search by ingredient adds friction to the information-seeking process. The lack of visual cues and intuitive design makes it hard for users to quickly identify key information such as overlapping ingredients or

safe dosage ranges. For individuals with lower health literacy, this creates a significant barrier to safe self-medication.

2.4 Project Response to Existing Gaps

Although the current OTC labeling system is designed to convey essential safety information, it reveals several structural shortcomings in terms of readability, comprehension, and usability. In particular, the reliance on dense text without visual support makes the system inaccessible to many users, especially older adults and those with limited health literacy.

In response to these challenges, this project proposes a user-centered approach to OTC information delivery. Specifically, it introduces an interactive visualization tool that helps users explore ingredient data and dosage risks in a more intuitive way. The tool visually maps ingredient overlaps between products, outlines safe dosage thresholds, and flags potential risks when combining medications—making it easier for consumers to make safe, informed decisions without relying solely on complex text-based materials. This approach—grounded in visual clarity and user-centered design—serves as the foundation for the interactive tool developed in this project. In the following section, I detail how this system was conceptualized, structured, and implemented.

3. Project Design & Implementation

3.1 Project Design Overview

This project was designed with the goal of making the informational risks of over-the-counter (OTC) medications—particularly overlapping ingredients and improper dosing—more visible and understandable. Rather than simply presenting medical facts in a static or list-based

format, the approach focused on user-centered interactivity: enabling people to actively explore, compare, and reflect on the OTC products they use every day.

The initial idea emerged from observing how people choose medications in everyday situations, whether at home or in the pharmacy aisle. In many cases, users are driven by brand familiarity or visual packaging rather than a true understanding of the product's contents. I wanted to replicate and challenge this behavioral moment by offering an interface that doesn't just present content, but reshapes the experience of reading, interpreting, and choosing medications.

To start, I compiled a list of commonly used OTC medications across categories like allergy, pain relief, cough and cold, digestive issues, arthritis, and fever. I then collected and categorized each product by brand name, active ingredients, dosage, and packaging. Using authoritative resources such as WebMD, Drugs.com, RxList, DailyMed, GoodRx, and HealthInHand.org, I built a structured dataset that mapped each drug to its core chemical components and potential risks. Additional ingredient-specific data—including toxicity, maximum daily dosage, and time intervals between doses—were sourced from the FDA and NIH.

On the visual and conceptual side, the system is grounded in ingredient-first thinking. Rather than guiding users through brand names, the interface emphasizes what's inside the medicine. To accomplish this, I designed a series of interconnected modules that gradually lead users from surface-level familiarity (such as package recognition) to deeper, more informed interactions based on shared ingredients and risk awareness.

The interface was developed using Vue.js for structure and D3.js for dynamic, data-driven visuals. JSON files containing structured information (ingredients, brands, dosages, and warnings) are parsed and used to generate real-time visualizations. When users interact with different combinations of products, the system responds immediately to highlight duplicate ingredients or display relevant warnings—creating a feedback loop that fosters behavior change.

Ultimately, the goal of the project was not simply to improve access to information, but to redesign the cognitive process of medication decision-making. The visual format and exploratory structure are intended to help users slow down, make sense of the risks, and build a more conscious relationship with the medications they consume.

3.2 Interface Architecture

The interactive tool is implemented as a single-page web application built with Vue.js and enhanced with D3.js for visual rendering. The overall layout is divided into six sequential sections: Problem, Ingredients, Categories, Try, Dosage Guide, and Label Check. These sections are navigable through a fixed navigation bar and are structured to follow a logical narrative arc—guiding users from the recognition of an issue to taking more informed action.

Each section is self-contained but contextually linked. The visual continuity between sections helps maintain user engagement while supporting a progressive learning journey. The aim is to simulate the kind of decision-making flow a consumer might experience in real life—from encountering a symptom, to picking up a familiar package, to wondering whether it's safe to combine with something else they've already taken.

3.3 Section-by-Section Breakdown

(1) Problem

The opening section introduces the project's core concern: the majority of Americans regularly use OTC drugs, but few consult healthcare providers or read Drug Facts labels. This section delivers that insight through bold typography and an attention-catching statistic: "9 out of 10 use OTCs; 8 out of 10 do so without medical advice." The design uses color contrast and direct questions ("Have you read the label?") to create a cognitive hook that sets the tone for the rest of the experience.

(2) Ingredients

Here, users are shown a grid of 73 commonly used OTC products organized by symptom category. Each item is represented by its package image to tap into visual familiarity. When users click a package, the card flips to reveal its active ingredients, which are tagged with color-coded labels. Ingredients are assigned consistent colors across all products, allowing users to visually track which compounds appear repeatedly in different medications.

This section is designed to shift the user's frame of reference from branding to ingredients. As users explore multiple products, repeated exposure to ingredient labels enables pattern recognition—an important cognitive step toward safer self-medication.

(3) Categories

The next section builds on ingredient awareness by highlighting specific high-risk compounds such as acetaminophen and diphenhydramine. These ingredients are displayed using a D3-based arc diagram, where each is linked to the types of symptoms or drug

categories it appears in. Clicking on an ingredient reveals a brief warning or explanation of associated health risks.

This format encourages reflection by showing how a single ingredient may appear in multiple contexts. It helps demystify chemical names and connects risks to real-world usage scenarios, building a more informed understanding of what each ingredient does—and why it matters.

(4) Try

The "Try" section lets users simulate combining up to three OTC medications. When selected medications share an ingredient, a visual warning appears, identifying the ingredient and advising the user to check dosage limits. The tool also prevents users from selecting more than three products at once, nudging them to focus on manageable decisions.

This interactivity mirrors real-life situations where people take multiple drugs for co-occurring symptoms—cough and headache, for example—without realizing they may be doubling up on a compound like acetaminophen. By surfacing this overlap, the section aims to reduce accidental overdosing.

(5) Dosage Guide

In this section, dosage information is presented using pill-shaped icons. Each ingredient is visualized with a horizontal row of pills: darker ones for a typical single dose, lighter ones for the maximum daily limit. The visual language of size, color, and spacing makes it easier for users to grasp concepts like "don't exceed 8 pills per day" than reading fine print on a box.

Additionally, recommended intervals between doses (in hours) are displayed to reinforce the importance of spacing. By presenting dosage data visually rather than numerically, the tool reduces cognitive friction and supports better recall.

(6) Label Check

The final section returns to the Drug Facts label, showing a real OTC label image with interactive hotspots. Users can hover or click to see explanations of each section: active ingredients, usage, warnings, directions, and other details. This reinforces earlier insights and closes the loop by encouraging label literacy as a long-term behavior.

3.4 Content and Data Collection

The data used in this project was meticulously gathered and cross-verified from multiple trusted health information platforms, including WebMD, Drugs.com, RxList, GoodRx, DailyMed, and HealthInHand.org. These sources provided comprehensive details on active ingredients, dosages, usage categories, and product-specific warnings.

All ingredient risk data—such as overdose potential, toxicity symptoms, and FDA guidelines for maximum usage—was taken from NIH and FDA publications. To enhance reliability, each data point was manually cross-checked across at least two platforms. Packaging images were sourced under open licenses or retrieved from official brand websites, ensuring visual consistency and legal clarity.

3.5 Narrative Structure and Design Strategy

The design strategy for this project is grounded in a progressive narrative framework: from problem recognition to active decision-making. Each section was built to represent a specific cognitive step:

Problem awareness → Ingredient recognition → Risk understanding → Combination simulation → Safe usage guidance → Label literacy.

By mirroring the user's mental model of OTC use, the interface helps shift their focus from visual branding to chemical content. The interactivity is not just functional, but pedagogical—it teaches users to notice, compare, and act more thoughtfully.

Crucially, the tool prioritizes visual clarity to support users with lower health literacy. Instead of relying on blocks of text or medical jargon, it uses familiar imagery, bold warnings, color coding, and interactive feedback to reduce cognitive load. The intention is not only to inform, but to reframe the user's relationship with OTC medications—turning every click into a moment of learning and potentially a safer choice.

4. Analysis and Discussion

4.1 Visualization Patterns and Ingredient Overlap

The interactive visualization tool developed in this project proved effective in structurally revealing the often-overlooked issue of overlapping ingredients among commonly used over-the-counter (OTC) medications in the United States. Through visual mapping, it became immediately apparent that key active ingredients—such as acetaminophen, diphenhydramine, and dextromethorphan—frequently appear across products targeting different symptoms,

including pain, cough, allergies, and fever. While consumers typically rely on brand logos and packaging in choosing medications, this tool invites them to notice the recurrence of ingredient names and colors, prompting a moment of recognition—“Wait, isn’t that the same ingredient again?” Such realizations are rarely prompted by traditional Drug Facts labels, which are often ignored or misunderstood. This tool, in contrast, transforms that obscured information into a clear, user-centered experience that supports safer decision-making, especially when multiple medications are involved.

4.2 Behavioral Shift and Ingredient-Centric Thinking

The arc diagram in the Categories section plays a particularly important role in shifting users' attention from branding to substance. By visually linking specific ingredients to the categories of symptoms they treat, the tool helps users recognize patterns in how certain ingredients are commonly repurposed across different OTC products. Instead of navigating medications by commercial names, users are encouraged to evaluate products based on efficacy and potential risk—fostering a more informed and critical mindset. The Try section takes this further by allowing users to simulate combining up to three OTC drugs. If the selected medications contain overlapping ingredients, a warning is triggered alongside a message reminding the user to check dosage limits. This interactivity pushes the tool beyond simple data delivery; it actively intervenes in user behavior and promotes safer medication practices.

4.3 Design Ethics and Communication Challenges

During the design process, one of the central challenges was maintaining a balance between minimizing information overload and ensuring comprehensive, accurate communication. Text-heavy explanations were deliberately avoided in favor of visual language—colors,

icons, shapes—that can convey complex information more intuitively. However, such visual simplification also carries risks of omitting nuance or reducing clarity for users who seek more detailed information. To address this, the interface subtly encourages users to treat the tool as a supplement rather than a substitute: a supportive entry point, not a replacement for medical advice. The final section, which revisits the Drug Facts label, reinforces this by helping users learn to interpret official labels more confidently in future real-world scenarios.

Overall, this project goes beyond merely presenting pharmaceutical data. It seeks to reframe how consumers think about OTC medications and redesigns the decision-making process itself. By inviting users to see past surface-level packaging and focus instead on what’s inside—and how much is too much—it becomes a case study in design-driven health literacy intervention. The tool holds particular promise for users with low health literacy, for whom traditional medical texts can be exclusionary. Ultimately, the work shows that altering the structure and format of health information can influence not only what users learn but how they act.

5. Conclusion

5.1 Summary and Contributions

This project was developed in response to a growing public health concern: the risks associated with self-directed use of over-the-counter (OTC) medications, particularly due to overlapping active ingredients and improper dosing. Although OTC drugs are widely perceived as simple and safe—reinforced by their availability across pharmacies, grocery stores, hotel pantries, and online retailers—the information required to use them safely is often complex, inconsistently communicated, and difficult to access. This disconnect between

accessibility and understanding became the central design problem this thesis sought to address.

Rather than presenting OTC drug information through traditional means such as text-heavy Drug Facts labels or static medical websites, this project proposes a user-centered, visualization-based approach. The core contribution is an interactive tool that enables users to visually explore ingredients, simulate drug combinations, understand dosage limits, and learn how to read medication labels—all within a guided, narrative-driven interface. The project combines Vue.js and D3.js to build a responsive and exploratory environment where the user experience is grounded not in abstract numbers or technical language, but in intuitive design and contextual learning.

Through the interface's six sections—Problem, Ingredients, Categories, Try, Dosage Guide, and Label Check—the user is gradually shifted from a brand-centric to a substance-centric perspective. At the start, users interact with familiar brand imagery, which then reveals ingredient data through flipped cards and color-coded indicators. This subtle shift encourages recognition of active ingredients over brand recognition, leading to deeper awareness of potential ingredient overlap. In the Try section, users can simulate combining up to three products and receive real-time feedback when overlaps occur, accompanied by visual warnings and dosage reminders. This design not only conveys information but nudges users toward safer behavioral choices.

The use of arc diagrams, pill-shaped dosage graphics, and layered ingredient-color associations further supports a visual language of risk communication. Importantly, the tool avoids information overload by prioritizing clarity and interaction over raw data density. Every interaction—from flipping a card to highlighting overlapping ingredients—serves as

both an educational moment and a design intervention aimed at reshaping how users think about everyday medications.

This thesis, therefore, offers not just a functional prototype but a conceptual framework for how public health tools might better align with the ways real people make decisions. It reflects an attempt to reframe health literacy not as a static body of knowledge, but as an ongoing, interactive practice—one that design can meaningfully support. Especially for users with lower health literacy, the interface reduces cognitive friction and empowers independent, informed decision-making. In this way, the project stands as a model for how data visualization can function as both a communication tool and a form of behavioral guidance within healthcare contexts.

5.2 Future Directions

While the current prototype demonstrates proof-of-concept, there are numerous pathways for further development and meaningful expansion.

First, expanding the dataset to include a broader range of OTC medications would be a significant next step. Currently, the tool includes approximately 70 commonly used OTC drugs in the U.S., but the actual market consists of thousands of products, many of which vary by region, age group, and consumer preference. A larger dataset would allow the tool to be more representative and better tailored to different populations and contexts of use.

Second, the system could evolve to analyze not just duplicated ingredients, but pharmacological interactions between drugs—whether they are contraindicated, synergistic, or neutral when taken together. Currently, users are warned when active ingredients overlap, but with the inclusion of interaction data, the system could offer even more nuanced safety

guidance. For example, the tool could flag combinations that increase drowsiness or liver toxicity, even if the ingredients are technically different.

Third, an especially promising direction is the integration of prescription drug data alongside OTC products. Many users regularly take prescription medications and are unaware of how they might interact with OTC options. Allowing users to input their current prescriptions and receive interaction alerts would elevate the tool from an OTC reference into a more comprehensive personal medication assistant.

Fourth, accessibility could be greatly enhanced through photo-based machine learning. A future version of the tool might allow users to upload a photo of the medications they keep at home—such as a picture of pill bottles or boxes in a drawer—and automatically detect the active ingredients and risks. This would be particularly useful for older adults, non-native English speakers, or individuals with limited digital literacy, as it removes the need to type or search manually.

Fifth, the interface could be further optimized for mobile use and expanded into multiple languages. Given that many OTC purchases and health-related searches now occur on smartphones, making the tool mobile-friendly would significantly increase its usability. Multilingual support—particularly in Spanish, Mandarin, Korean, and other widely spoken languages—would expand its reach to immigrant and multilingual communities, who often face additional barriers to health information.

Finally, a more systemic vision involves integrating this kind of tool into the point-of-sale experience itself. One future possibility is to embed QR codes on OTC packaging, which consumers could scan to instantly access ingredient information, safe combinations, and dosage instructions. This would make visualized medication safety guidance available at the exact moment of decision-making, essentially bringing design into the pharmacy shelf.

These expansion ideas are not merely technical enhancements—they point toward a broader mission of democratizing access to medication safety. As healthcare systems increasingly shift responsibility onto the individual, the importance of equitable, intuitive, and user-friendly information tools becomes more critical. This project illustrates how thoughtful design can intervene in that gap, supporting people not just with data, but with clarity, confidence, and choice.

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